Conformal Inference Assignment 3
(Deadline: November 1)

Wen, Zehai

Temps de compilation : 2026-10-16 17:57:54-04:00

Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 9.1

Consider full conformal prediction with an exchangeable dataset and a symmetric score function. Instead of comparing the test score against all n training scores, compare it against a random subset indexed by $I \subseteq \{1, \ldots, n\}$, sampled independently of the data:

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} : S_{n+1}^y \leqslant \text{Quantile}\left((S_i^y)_{i \in I}; (1-\alpha)\left(1 + \frac{1}{|I|}\right)\right)\right\}.$$

Prove that this modified prediction set has marginal coverage at least $1 - \alpha$.

Solution Let $Z_i = (X_i, Y_i)$, $D_{n+1} = (Z_1, \ldots, Z_{n+1})$, and define the scores at the true test response by

$$S_i = s(Z_i; D_{n+1}), \qquad i = 1, \ldots, n+1.$$

Symmetry of s and exchangeability of the data imply that $(S_1, \ldots, S_{n+1})$ is exchangeable. Indeed, for every permutation π of the $n+1$ indices,

$$(S_{\pi(i)})_{i=1}^{n+1} = \big(s(Z_{\pi(i)}; (D_{n+1})_\pi)\big)_{i=1}^{n+1} \overset{d}{=} (S_i)_{i=1}^{n+1}.$$

The equality uses symmetry in the dataset argument, and the equality in distribution uses exchangeability of the data.

Now condition on $I = J$, for any fixed nonempty subset J having positive probability, and put $m = |J|$. Independence of I and the data means that conditioning on $I = J$ does not change the distribution of the scores. Thus the m selected scores together with S_{n+1} remain exchangeable. Let

$$k_m = \lceil (m+1)(1-\alpha) \rceil.$$

If $k_m \leqslant m$, the quantile in the question is the k_mth smallest score among $(S_i)_{i \in J}$. The exchangeable-rank argument gives

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid I = J\} \geqslant \frac{k_m}{m+1} \geqslant 1 - \alpha.$$

For completeness, break any score ties uniformly at random solely for this rank argument. The rank of S_{n+1} among the $m+1$ scores is then uniform on $\{1, \ldots, m+1\}$. Rank at most k_m implies that S_{n+1} is no larger than the selected-sample quantile, which proves the first inequality even when scores tie. If $k_m = m+1$, the quantile is $+\infty$ by the conformal convention, and the conditional coverage is 1.

Averaging the conditional guarantees over all possible subsets yields

$$\boxed{P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = \sum_J P(I = J) P\{Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid I = J\} \geqslant 1 - \alpha.}$$

If an empty subset is allowed, define its threshold to be $+\infty$ and return $\mathcal{Y}$ on that event; the same conclusion follows. Notice that I need not be uniform among subsets, nor have a fixed size. ////

Exercise 9.2

For $\mathcal{Y} = \mathbb{R}$, consider K-nearest-neighbor regression,

$$\hat{f}(x; D) = \frac{1}{K} \sum_{j \in N(x,D)} y_j,$$

where $N(x, D)$ indexes the K nearest feature vectors to x in D. Show that this algorithm has the affine dependence on the candidate response required by (9.4), so that its full conformal prediction set can be computed efficiently.

Solution Fix the training data $D_n = ((X_j, Y_j))_{j=1}^n$ and the test feature X_{n+1}. For a candidate response y, let

$$D_{n+1}^y = ((X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)).$$

Use a fixed rule for distance ties that depends only on features and indices, rather than response values. Then the neighbor set $N(x', D_{n+1}^y)$ does not depend on y: the augmented feature vectors are the same for every candidate. Denote this set by $N_{x'}$. Separating out the possible contribution of the test point gives

$$\begin{aligned} \hat{f}(x'; D_{n+1}^y) &= \frac{1}{K} \sum_{j \in N_{x'} \cap \{1,\ldots,n\}} Y_j + \frac{\mathbf{1}\{n+1 \in N_{x'}\}}{K}\, y \\ &= a(x'; D_n, X_{n+1}) + b(x'; D_n, X_{n+1})\, y, \end{aligned}$$

where

$$\boxed{a(x'; D_n, X_{n+1}) = \frac{1}{K} \sum_{j \in N_{x'} \cap \{1,\ldots,n\}} Y_j, \qquad b(x'; D_n, X_{n+1}) = \frac{\mathbf{1}\{n+1 \in N_{x'}\}}{K}.}$$

Both coefficients are independent of y, establishing (9.4).

To see explicitly why this helps, write $a_i = a(X_i; D_n, X_{n+1})$ and $b_i = b(X_i; D_n, X_{n+1})$. For the absolute residual score, the training and test scores are

$$S_i^y = |Y_i - a_i - b_i y| \quad (i \leqslant n), \qquad S_{n+1}^y = |(1 - b_{n+1})y - a_{n+1}|.$$

Thus

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathbb{R} : \frac{1 + \sum_{i=1}^n \mathbf{1}\{|Y_i - a_i - b_i y| \geqslant |(1 - b_{n+1})y - a_{n+1}|\}}{n+1} > \alpha \right\}.$$

Each comparison can change only at a solution of one of the two linear equations

$$Y_i - a_i - b_i y = \pm\big((1 - b_{n+1})y - a_{n+1}\big).$$

Equations that hold identically contribute no boundary; each remaining equation has at most one solution. After computing the neighbor sets and coefficients once, there are therefore at most $2n$ candidate boundary points. Evaluate the p-value between consecutive boundary points and at the boundary points themselves to recover the exact prediction set. ////

Exercise 9.3

Fix $n \geqslant 1$. Consider the naive discretization strategy (9.6), using a finite uniform grid $G = \{y^{(1)}, \ldots, y^{(M)}\}$ over $[-B, B]$. Construct a distribution P and a symmetric score function s such that this approximation has zero marginal coverage for every choice of grid size M and range $B > 0$.

Solution First consider the nontrivial range $\alpha \geqslant 1/(n+1)$, with $0 < \alpha < 1$. The very small α case is addressed below. Take $\mathcal{X} = \mathcal{Y} = \mathbb{R}$ and define P by

$$X \sim \mathrm{Unif}(0, 1), \qquad Y = X.$$

Sample (X_i, Y_i) independently from P and choose

$$\boxed{s((x, y); D) = |y - x|.}$$

This score is symmetric because it does not depend on D at all.

For every candidate response y, all n training scores are zero:

$$S_i^y = |Y_i - X_i| = 0, \qquad i = 1, \ldots, n,$$

whereas the test score is $S_{n+1}^y = |y - X_{n+1}|$. Since $k = \lceil (n+1)(1-\alpha) \rceil \leqslant n$, the conformal quantile of the training scores is zero. Consequently, the exact full conformal prediction set is

$$\mathcal{C}_{\text{full}}(X_{n+1}) = \{X_{n+1}\},$$

which covers $Y_{n+1} = X_{n+1}$ with probability 1.

Now fix any finite grid G. Since X_{n+1} is continuously distributed,

$$P\{X_{n+1} \in G\} \leqslant \sum_{m=1}^{M} P\{X_{n+1} = y^{(m)}\} = 0.$$

Therefore, almost surely every grid point has strictly positive test score and fails the comparison against the zero calibration quantile. The accepted grid-index set is empty. In particular, the expansion of accepted grid points into intervals in (9.6) produces

$$\mathcal{C}_{\text{naive}}(X_{n+1}) = \bigcup_{m \in \varnothing} (y^{(m-1)}, y^{(m+1)}) = \varnothing \quad \text{almost surely.}$$

Hence

$$\boxed{P\{Y_{n+1} \in \mathcal{C}_{\text{naive}}(X_{n+1})\} = 0.}$$

The same P and s work for every fixed finite grid, including every uniform grid of any resolution or range. For $M = 1$, interpret the grid as any chosen single point; the argument is unchanged.

The necessary restriction on α As written, the exercise omits a restriction needed for its conclusion. If $0 < \alpha < 1/(n+1)$, then

$$\lceil (n+1)(1-\alpha) \rceil = n + 1.$$

The conformal quantile is then $+\infty$, so every grid point is accepted for any real-valued score function. With the boundary convention $y^{(0)} = -\infty$ and $y^{(M+1)} = +\infty$ from Section 9.2.3,

$$\mathcal{C}_{\text{naive}}(X_{n+1}) = \bigcup_{m=1}^{M} (y^{(m-1)}, y^{(m+1)}) = \mathbb{R}.$$

Its coverage is 1, making zero coverage impossible in this range. Thus the counterexample applies precisely when $\alpha \geqslant 1/(n+1)$, or equivalently when the conformal threshold is a finite calibration order statistic. ////

Exercise 9.4

Let A be a regression algorithm that takes an ordered dataset D of size m and returns $\hat{f}(\cdot; D)$, without assuming symmetry. Let $\xi \sim \mathrm{Unif}[0,1]$ and choose a map $\xi \mapsto \sigma_\xi \in \mathfrak{S}_m$ such that σ_ξ is uniform among all permutations. Define

$$s((x,y); D, \xi) = |y - \hat{f}(x; D_{\sigma_\xi})|.$$

Prove that s is a symmetric randomized score function in the sense of Definition 9.1.

Solution Fix any dataset $D = (z_1, \ldots, z_m)$ and any permutation $\pi \in \mathfrak{S}_m$, where $z_j = (x_j, y_j)$. If $\sigma \in \mathfrak{S}_m$, then

$$(D_\pi)_\sigma = (z_{\pi(\sigma(1))}, \ldots, z_{\pi(\sigma(m))}) = D_{\pi\circ\sigma}.$$

The map $\sigma \mapsto \pi\circ\sigma$ is a bijection of $\mathfrak{S}_m$, with inverse $\rho \mapsto \pi^{-1}\circ\rho$. Thus composing a uniform random permutation with the fixed permutation π preserves its uniform distribution:

$$\pi \circ \sigma_\xi \overset{d}{=} \sigma_\xi.$$

In particular, shuffling D_π uniformly has the same distribution as shuffling D uniformly.

To verify the full requirement of Definition 9.1, fix any finite list of evaluation points $(x'_1, y'_1), \ldots, (x'_r, y'_r)$ and define

$$F(\rho) = \big(|y'_j - \hat{f}(x'_j; D_\rho)|\big)_{j=1}^r, \qquad \rho \in \mathfrak{S}_m.$$

Using the same random seed for all r scores, we obtain

$$\begin{aligned}\big(s((x'_j, y'_j); D_\pi, \xi)\big)_{j=1}^r &= F(\pi \circ \sigma_\xi) \\ &\overset{d}{=} F(\sigma_\xi) \\ &= \big(s((x'_j, y'_j); D, \xi)\big)_{j=1}^r.\end{aligned}$$

This proves equality in distribution of the entire score vector for arbitrary evaluation points and any permutation of D, exactly as required by Definition 9.1. No symmetry of A is needed: the random shuffle gives every possible ordering of the data the same probability. ////